

SYNTHETIC DEMONSTRATION — every figure on these pages was produced from a synthetic genome. No personal genomic, clinical, or laboratory data is present.

ECONOMIC INTELLIGENCE FOR GENOME SEQUENCING

A sequenced genome is evidence. GenomeLens turns it into a decision, and prices the decision.

GenomeLens is an economic intelligence layer for genome sequencing. It turns sequence data into measurable decision and economic value. It reads a variant callset and returns the clinical decisions that callset can change, who benefits from each one, what each is worth, and — explicitly — which ones it refuses to price.

WHAT THE MODEL RECOVERED FROM ONE GENOME

12

Clinical pathways carried through the economic model

7

Distinct economic mechanisms evaluated

4

Distinct beneficiaries the value is separated across

9

Pathways blocked on a named evidence gap

THE VALUE, SEPARATED BY KIND AND BY WHO RECEIVES IT

Modeled payoff if the triggering circumstances occur

\$110,766

SPANS MULTIPLE BENEFICIARIES — NOT ONE PERSON'S GAIN

Modeled potential economic benefit across supported findings if the relevant medication, reproductive, or clinical circumstances occur. Not all identified value is expected to be realized, and it does not all accrue to the same person.

Expected present value, sequenced individual

\$112.28

ACCRUES TO THE SEQUENCED INDIVIDUAL

Probability-weighted and discounted. Lower-bound scenario: pathways blocked on an unsourced parameter contribute nothing.

Expected present value, payer

\$4.55

ACCRUES TO THE PAYER

Testing the payer may not need to repeat. Claimed only where the existing result is an adequate substitute for the test it would replace.

These three numbers are not comparable and are not additive. The first is a payoff conditional on an event that has not happened, pooled across people who are not the same person. The second and third are probability-weighted present values, each scoped to a single beneficiary. Color encodes which kind a number is, never how large it is. For scale, \$183.16 of management value is modeled in total — the worth of the correct treatment however it is reached — of which \$112.28 is attributed to the genome. GenomeLens books the attributable share, never the whole.

HOW A PAYOFF BECOMES A VALUE

Worked example — DPYD / fluoropyrimidines

$$\begin{array}{ccccccc} \text{DPYD} & \rightarrow & \$11,600 & \times & 1.05\% & = & \$38.91 \\ \text{identity verified} & & \text{payoff if the drug} & & \text{probability that decision} & & \text{expected present value} \\ & & \text{decision arises} & & \text{arises} & & \end{array}$$

Reduced DPD activity can substantially increase fluoropyrimidine toxicity risk. Genotype-guided dose reduction or alternative therapy.

Both numbers are true and they are not interchangeable. A platform that reports the first as value delivered is overstating its product by roughly two orders of magnitude. GenomeLens reports both, labeled, and lets the buyer choose the one their question needs.

What GenomeLens refuses to price is the reason to trust what it does price.

Across 7 economic mechanisms evaluated on this genome, 3 resolved to a defensible number and 9 pathways are blocked on a named evidence gap. 6 of those still carry a conditional payoff that is reported in the headline; what is blocked is the step that would turn it into a bookable expected present value. Each names the parameter, identity, or attribution problem responsible.

VALUE SEPARATED BY MECHANISM AND BENEFICIARY

MECHANISM	BENEFICIARY	PATHWAYS	CONDITIONAL PAYOFF	STATE
individual health	accrues to the sequenced individual	1	—	Attribution unresolved
pharmacogenomic	accrues to the sequenced individual	4	\$52,150	Conditional
diagnostic odyssey	accrues to the sequenced individual	1	—	Attribution unresolved
replacement testing	accrues to the payer	4	\$519.52	Conditional
reproductive	accrues to a prospective child	1	\$58,097	Conditional
family cascade	accrues to a relative	1	—	Attribution unresolved
future option	accrues to the sequenced individual	0	—	Not currently modeled

Beneficiary separation is structural, not editorial. A payer does not receive reproductive value and an individual does not receive cascade value, so the model never pools them into a single figure attributed to one party.

PATHWAYS BLOCKED ON A NAMED EVIDENCE GAP

- **reproductive — Attribution unresolved.** P(planning a pregnancy) is not sourced *\$58,097 conditional payoff reported; the expected present value is what is blocked.*
- **pharmacogenomic — Parameter blocked.** P(abacavir consideration | HIV diagnosis) [unsourced] *\$36,425 conditional payoff reported; the expected present value is what is blocked.*
- **pharmacogenomic — Parameter blocked.** P(an indication for thiopurine therapy (IBD, autoimmune disease, or ALL maintenance)) [unsourced]; P(thiopurine prescription | an... *\$15,725 conditional payoff reported; the expected present value is what is blocked.*
- **replacement testing — Identity blocked.** NOT a valid substitute. Three of the four CPIC-tested DPYD alleles are identity-verified here (a tested allele, a tested allele... *\$174.81 conditional payoff reported; the expected present value is what is blocked.*
- **replacement testing — Parameter blocked.** P(an indication for thiopurine therapy (IBD, autoimmune disease, or ALL maintenance)) [unsourced]; P(thiopurine prescription | an... *\$174.81 conditional payoff reported; the expected present value is what is blocked.*
- **replacement testing — Identity blocked.** NOT a valid substitute. This pipeline reaches HLA-B*57:01 by TAG PROXY rather than direct typing, and an internal consistency... *\$169.90 conditional payoff reported; the expected present value is what is blocked.*

3 further blocked pathways are itemized in the detailed economic report. These are the least material blockers, not the least defensible ones.

WHAT A PARTNER SHOULD TAKE FROM THIS

Realized value is reported honestly. Value realized to date on this genome is \$0, because nothing has triggered yet. A platform that cannot show a zero is not measuring realization.

The arithmetic reconciles. Every figure traces to a named parameter with a source, and cross-arm totals are checked against the headline at build time. The detailed economic report carries the full ledger, the model framework, the validation checks, and the declared limitations.